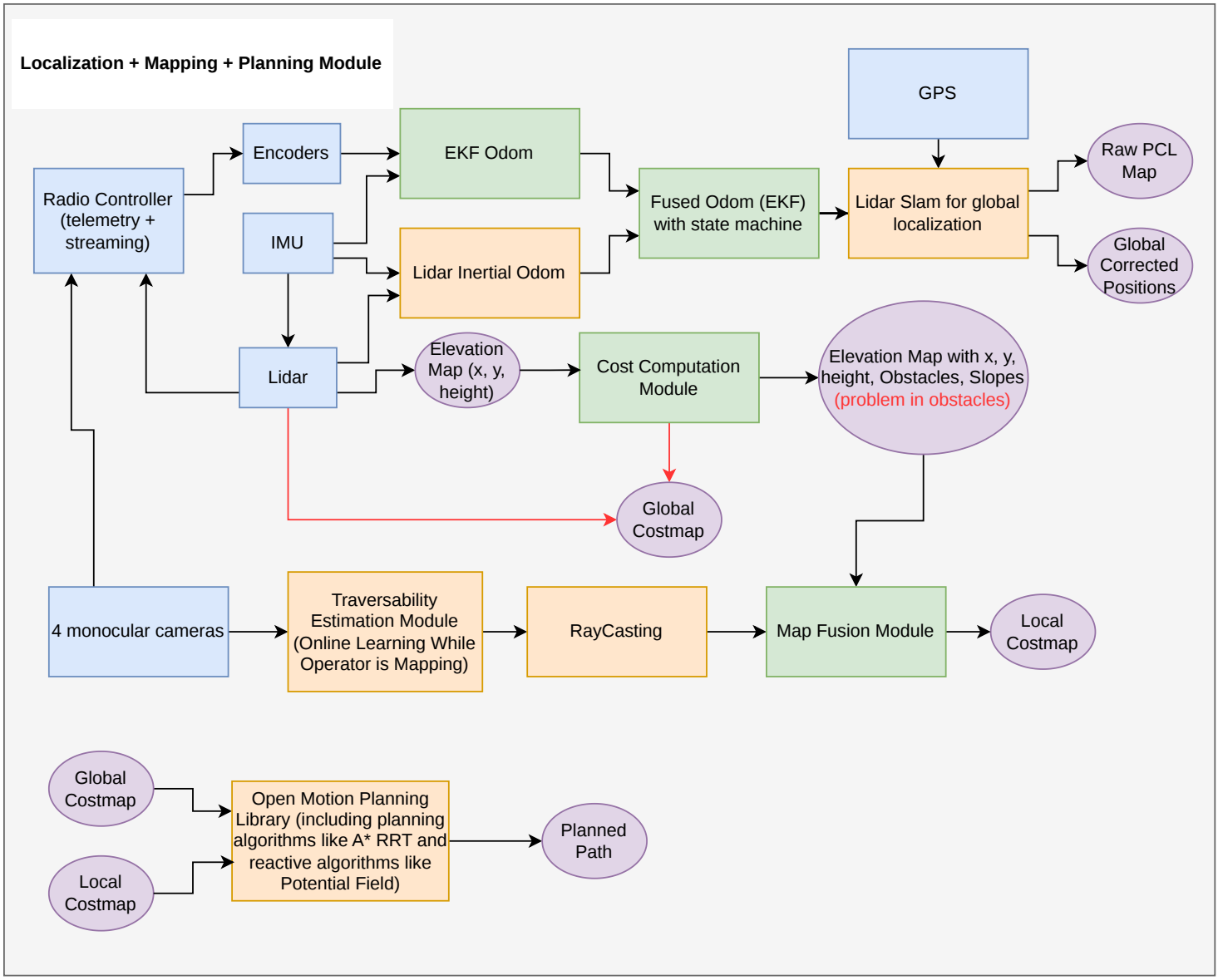
Localization + Mapping + Planning Module
GPS
Encoders
EKF Odom
Raw PCL Map
Radio Controller (telemetry + streaming)
Fused Odom (EKF) with state machine
Lidar Slam for global localization
IMU
Lidar Inertial Odom
Global Corrected Positions
Lidar
Elevation Map (x, y, height)
Cost Computation Module
Elevation Map with x, y, height, Obstacles, Slopes (problem in obstacles)
Global Costmap
4 monocular cameras
Traversability Estimation Module (Online Learning While Operator is Mapping)
RayCasting
Map Fusion Module
Local Costmap
Global Costmap
Open Motion Planning Library (including planning algorithms like A* RRT and reactive algorithms like Potential Field)
Planned Path
Local Costmap